Code-Layout, Readability and Reusability

Date	02 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

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Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform 4-space indentation used throughout the project.	**Yes**	Follows PEP 8 coding standards.
2	Proper File Structure	Organized into Dataset, Flask (templates, static), and Training folders.	**Yes**	Clean and modular project layout.
3	Meaningful Variable Names	Descriptive variable names such as `life_expectancy`, `gni_per_capita`, `prediction`.	**Yes**	Improves code readability.
4	Function / Method Names	Functions like `predict()`, `validate_inputs()`, and `classify_hdi()`.	**Yes**	Uses meaningful verb-based names.
5	Code Comments	Important modules and functions include comments explaining their purpose.	**Yes**	Enhances maintainability.
6	Modular Design	Data preprocessing, prediction, and validation are implemented separately.	**Yes**	Easy to modify and extend.
7	No Redundant Code	Avoids duplicate logic by reusing common functions.	**Yes**	Promotes clean code practices.
8	Error Handling	Validates user inputs and handles invalid requests gracefully.	**Yes**	Prevents application crashes.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	`classify_hdi()`	Python	Used for categorizing predicted HDI values.	High
2	`validate_inputs()`	Python / Flask	Used to validate user inputs before prediction.	High
3	`base.html`	HTML / Jinja2	Extended by Home, Predict, and About pages.	High
4	`style.css`	CSS3	Shared across all web pages.	High
5	`HDI.pkl`	Pickle / Scikit-learn	Loaded once and reused for every prediction.	High

Overall Code Quality Assessment:

Aspect	Rating	Comments
Code Layout & Structure	**5 / 5**	Well-organized folders with a clear project structure.
Readability	**5 / 5**	Uses meaningful variable names and consistent formatting.
Reusability	**5 / 5**	Common functions and templates are reused effectively.
Documentation / Comments	**4.5 / 5**	Adequate comments and project documentation provided.
Overall Score	**4.9 / 5**	Clean, maintainable, and production-ready educational ML project.